

Name: Cuong Dang

Course: ITAI-1371-Intro to Machine Learning

Date: June 09, 2026

MODULE 02: REFLECTIVE JOURNAL

L02 ML Development Tools

1. Learning About Vectorization with NumPy

Before this lab, when I wanted to calculate math values for many rows of data, I always thought I needed to write a for loop to check each row one by one. In Task 1, I saw vectorization in action when I used `np.mean()` and `np.std()`.

Instead of writing a long loop, NumPy calculated the average of all 150 rows of sepal lengths at the exact same time. This made me realize that vectorization treats data as one big mathematical block. It is much faster and uses less computer memory, which is very important for handling big datasets in machine learning.

2. Understanding Pandas DataFrames and Indexing

At first, a pandas DataFrame just looked like a normal Excel spreadsheet to me. However, learning about indexing changed how I think about data.

I initially confused `.loc` and `.iloc`, but I realized that one uses names and the other uses row numbers. My biggest breakthrough was learning boolean indexing, like `df[df['species'] == species]`. It is amazing that a single line of code can filter thousands of rows instantly without complex logic. I now understand that pandas helps us group and clean data easily so that our machine learning models can receive perfect information.

3. Using Visuals to Understand Data Patterns

I used to think that making charts with matplotlib was just for making projects look pretty at the very end. This lab completely changed my mind. Making the scatter plot showed me that visualization is a powerful tool to check our data before training a model.

When looking at the raw rows of numbers, everything looked the same. But on the chart, I instantly saw that the Setosa species forms a separate group. This means an AI model will easily recognize it. However, the Versicolor and Virginica species mix together on the chart. This taught me that looking at visuals helps us find problems early and tells us exactly which parts of the data our machine learning model will struggle with.

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4. Conclusion and GitHub

This lab taught me that data tools work together like a team: NumPy does fast math, pandas organizes the data, and matplotlib helps us see the patterns.

Also, learning about GitHub made me realize that saving and documenting code is a professional requirement, not just a personal backup. Moving forward, I will focus more on finding patterns in data visually before trying to write machine learning code.
